1. $X$ is discrete uniform random variable

$$R_X = \{-2, -1, 0, 1, 2\} \quad \& \quad Y = 3X + 2$$

To find pmf of $Y$.

Since $X$ is discrete and uniform.

$$P(X = x) = \frac{1}{5} = 0.2, \quad \forall \; x \in \{-2, -1, 0, 1, 2\}$$

To determine range of $Y$

$x = -2 \qquad Y = -4$

$x = -1 \qquad Y = -1$

$x = 0 \qquad Y = 2$

$x = 1 \qquad Y = 5$

$x = 2 \qquad Y = 8$

$\therefore$ Range of $Y$ is $R_Y = \{-4, -1, 2, 5, 8\}$

Here, the function $Y = 3X + 2$ is one to one mapping from $R_X$ to $R_Y$ i.e. each value of $Y = y$ corresponds to one value of $X = x$

$$\therefore \quad P(Y = y) = P(X = x) = \frac{1}{5}$$

pmf of $Y$

$$P(Y = y) = \begin{cases} 1/5 & , y \in \{-4, -1, 2, 5, 8\} \\ 0 & \text{otherwise} \end{cases}$$

Ans

(2) $X$ the number of customers waiting for bus.
$X$ is geometric r.v with parameter $p$.
such that pmf $P_X(k) = p(1-p)^k$ for every non negative integer $k$.
Bus can take $M$ passengers, for $M > 0$
& $Y$ is defined as passengers left behind.

So let's consider some cases

i) $X \leq M$, i.e no customer is left behind
$\therefore Y = 0$

& ii) $X > M$ so $Y = (X - M)$ no. of customers are left behind.

To find pmf of $Y$, $P(Y=0)$ when $X \leq M$

$$P(Y=0) = P(X \leq M) = \sum_{k=0}^{M} p(1-p)^k$$

$$= p \sum_{k=0}^{M} (1-p)^k$$

for finite geometric series, $\sum_{k=0}^{n} r^k = \frac{1-r^{n+1}}{1-r}$, here $1-p = r$

$$\therefore P(Y=0) = p \frac{1-(1-p)^{M+1}}{1-(1-p)}$$

$$= 1 - (1-p)^{M+1}$$

To find pmf of $Y$, $P(Y = X - M)$ i.e. when $X > M$
let $X - M = y$ where $y \in \{1, 2, 3 \ldots\}$
$\therefore X = y + M$

$\therefore \quad P(Y=y) = P(X = M+y) = p(1-p)^{M+y}$ for $y \geq 1$

$\therefore$ Combined pmf

$$P_Y(y) = \begin{cases} 1-(1-p)^{M+1} & \text{if } y = 0 \\ p(1-p)^{M+y} & y = 1, 2, 3 \ldots \\ 0 & \text{otherwise} \end{cases}$$

Ans

3 $X$ exponential random variable with parameter $\lambda$

$$f_X(x) = \lambda e^{-\lambda x} u(x)$$

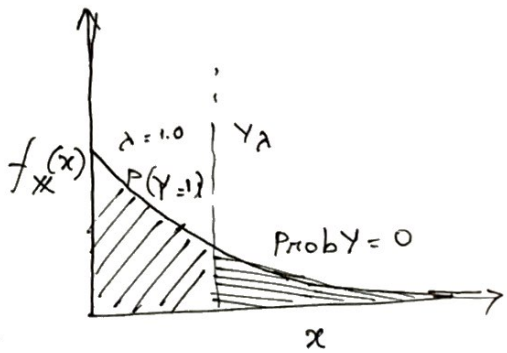
to find pmf of r.v

$$Y = \begin{cases} 1 & \text{if } X < 1/\lambda \\ 0 & \text{if } X \geq 1/\lambda \end{cases}$$

To find pmf of $Y$ we need to find prob of each outcome of $Y$, where $Y$ is a discrete r.v taking up values of 1 or 0.

$Y = 1$ if $X < 1/\lambda$, to find prob we integrate pdf of $X$ from 0 to $1/\lambda$

$$P(Y=1) = P\left(X < \frac{1}{\lambda}\right) = \int_{-\infty}^{1/\lambda} \lambda e^{-\lambda x} u(x)\, dx$$




$$= \int_{-\infty}^{0} \lambda e^{-\lambda x} u(x)\, dx + \int_{0}^{1/\lambda} \lambda e^{-\lambda x} u(x)\, dx$$

$u(x)$ is an indicator fn when $x < 0$, $u(x) = 0$ & when $x \geq 0$, $u(x) = 1$.

$$\therefore \quad P(Y=1) = \int_{-\infty}^{0} \lambda e^{-\lambda x} \cdot 0 \cdot dx + \lambda \int_{0}^{1/\lambda} e^{-\lambda x} \cdot 1 \cdot dx$$

or $P(Y=1) = P\left(X < \frac{1}{\lambda}\right) = \frac{\lambda}{\lambda}\left[-e^{-\lambda x}\right]_0^{1/\lambda} = \left[-e^{-\lambda \cdot \frac{1}{\lambda}} + e^0\right]$

$= \left[1 - \frac{1}{e}\right]$

now $P(Y=0) = P\left(X \geq \frac{1}{\lambda}\right) = \int_{1/\lambda}^{\infty} \lambda e^{-\lambda x} u(x)\, dx$

since $x \geq \frac{1}{\lambda}$ & $\frac{1}{\lambda} > 0$, $u(x) = 1$

$\therefore \int_{1/\lambda}^{\infty} \lambda e^{-\lambda x}\, dx = \left[-e^{-\lambda x}\right]_{1/\lambda}^{\infty}$

$= \left[0 + \frac{1}{e}\right] = \frac{1}{e}$

$\therefore$ final pmf

$$p_Y(y) = \begin{cases} 1 - \frac{1}{e} & , y = 1 \\ \frac{1}{e} & y = 0 \\ 0 & \text{otherwise} \end{cases}$$

Ans

4 $X$ & $Y$ are r.v defined on the prob space $(S, F, P)$

To show measures $P_X$ & $P_Y$ can be found

using $P_X(A) = P_{XY}(A \times \mathbb{R})$

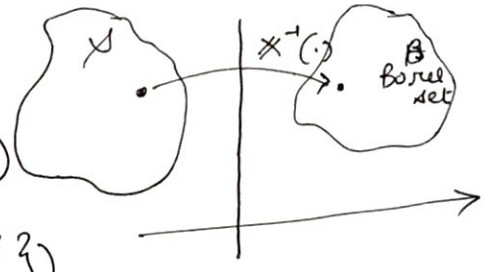
$P_Y(A) = P_{XY}(\mathbb{R} \times A)$ for any borel set $A$.

Given $X : S \to \mathbb{R}$ and $Y : S \to \mathbb{R}$ are random variables on $(S, F, P)$

$A \subset \mathbb{R}$

$\therefore P_X(A) = P(X^{-1}(A))$

$= P\{\omega \in S : X(\omega) \in A\}$




& $P_{XY}(A') = P((X,Y)^{-1}(A'))$ $A' \subset \mathcal{B}(\mathbb{R}^2)$

$= P(\{\omega \in S : (X(\omega), Y(\omega)) \in A'\})$

let us consider borel set $B = A \times \mathbb{R}$

$$P_{XY}(A \times \mathbb{R}) = P(\{\omega \in S : (X(\omega), Y(\omega)) \in A \times \mathbb{R}\})$$

$$(X(\omega), Y(\omega)) \in A \times \mathbb{R} \Rightarrow (X(\omega) \in A) \text{ and } (Y(\omega) \in \mathbb{R})$$

Also $Y$ r.v & $Y : S \to \mathbb{R}$ so $Y(\omega) \in \mathbb{R}$ for every $\omega \in S$

$\therefore \{\omega \in S : Y(\omega) \in \mathbb{R}\}$ is the sample space $S$.

$$\therefore P_{XY}(A \times \mathbb{R}) = P(\{\omega : X(\omega) \in A\} \cap \{\omega : Y(\omega) \in \mathbb{R}\}$$
$$= P(\{\omega : X(\omega) \in A\} \cap S\} = \{\omega : X(\omega) \in A\} = P_X(A)$$

for $P_{XY}(\mathbb{R} \times A)$ let Borel set $B = \mathbb{R} \times A$

$$P_{XY}(\mathbb{R} \times A) = P\{\omega \in S : (X(\omega), Y(\omega)) \in \mathbb{R} \times A\})$$

Same way as above,

$$(X(\omega), Y(\omega)) \in \mathbb{R} \times A \Rightarrow (X(\omega) \in A) \text{ and } (X(Y) \in A)$$

Since $X(\omega) \in \mathbb{R}$ is true for all $\omega \in S$:

$$S \cap \{\omega : Y(\omega) \in A\} = \{\omega : Y(\omega) \in A\}$$

$$\therefore P_{XY}(\mathbb{R} \times A) = P\{(\omega : Y(\omega) \in A\}$$
$$= P_Y(A)$$

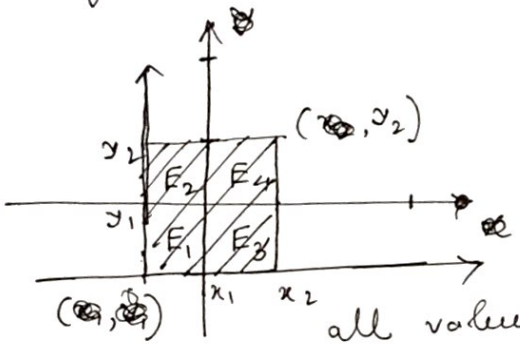
Ans.

5. joint cdf of r.v $X$ and $Y$ be $F_{XY}(x,y)$

To show,

$$P(\{x_1 < X \leq x_2\} \cap \{y_1 < Y \leq y_2\}) = F_{XY}(x_2, y_2) - F_{XY}(x_1, y_2) - F_{XY}(x_2, y_1) + F_{XY}(x_1, y_1) \quad (1)$$

for $x_1, x_2, y_1, y_2 \in \mathbb{R}$, $x_1 < x_2$, $y_1 < y_2$




Let's define two events A & B defined as $A = \{X \leq x\}$ and $B = \{Y \leq y\}$

These two events are independent for all values of $x$ & $y$, then we can say r.v $X$ & $Y$ are also independent

so $P\{X \leq x, Y \leq y\} = P\{X \leq x\} P\{Y \leq y\}$

Since A & B are independent, $P(A \cap B) = P(A) \cdot P(B)$.

or $F_{XY}(x,y) = F_X(x) \cdot F_Y(y)$ by definitions of cdfs.

differentiating both sides, w.r.t $x$ & $y$,

$$f_{X,Y}(x,y) = \frac{\partial^2}{\partial x \partial y}\left[F_X(x) F_Y(y)\right]$$

diff$^n$ w.r.t $x$

$$\frac{\partial}{\partial x}\left[F_X(x) F_Y(y)\right] = F_Y(y) f_X(x)$$

then diff w.r.t $y$

$$\frac{\partial}{\partial y}\left[F_Y(y) f_X(x)\right] = f_X(x) f_Y(y)$$

$$\therefore \quad f_{X,Y}(x,y) = f_X(x) f_Y(y) \text{ for all } x \text{ \& } y$$

RHS of ①

$= F_{XY}(x_2, y_2) - F_{XY}(x_1, y_2) - F_{XY}(x_2, y_1) + F_{XY}(x_1, y_1)$

$= F_X(x_2) F_Y(y_2) - F_X(x_1) F_Y(y_2) - F_X(x_2) F_Y(y_1) + F_X(x_1) F_Y(y_1)$

if $X$ and $Y$ are independent, then any interval events are also independent

$= F_Y(y_2)\left[F_X(x_2) - F_X(x_1)\right] - F_Y(y_1)\left[F_X(x_2) - F_X(x_1)\right]$

$= \left[F_Y(y_2) - F_Y(y_1)\right]\left[F_X(x_2) - F_X(x_1)\right]$

[Method found in ece.uwic.ca/~mmcguire/teaching/e400/Lecture9.pdf.]

$= P(x_1 < X \leq x_2)\, P(y_1 < Y < y_2)$

$= P(\{x_1 < X \leq x_2\} \cap \{y_1 < Y \leq y_2\})$ Ans

Now let's try to prove it the way purdue wants :

using the union of disjoint events

Goal is to prove

$P(\{x_1 \leq X \leq x_2\} \cap \{y_1 < Y \leq y_2\}) = F_{XY}(x_2, y_2) - F_{XY}(x_1, y_2) - F_{XY}(x_2, y_1) + F_{XY}(x_1, y_1)$

let's define event $A = \{X \leq x_2, Y \leq y_2\}$

we split $A$ in four disjoint regions with no overlap

$$A = E_1 \cup E_2 \cup E_3 \cup E_4.$$

$E_1 = \{X \leq x_1, Y \leq y_1\}$

$E_2 = \{X \leq x_1, y_1 < Y \leq y_2\}$

$E_3 = \{x_1 < X \leq x_2, Y \leq y_1\}$

$E_4 = \{x_1 < X \leq x_2, y_1 < Y \leq y_2\}$

we know from prob axiom 3

$$P(A) = P(E_1) + P(E_2) + P(E_3) + P(E_4)$$

$$P(E_4) = P(A) - P(E_1) - P(E_2) - P(E_3)$$

$$= F_{XY}(x_2, y_2) - F_{XY}(x_1, y_1) - P(E_2) - P(E_3)$$

Now $P(E_2) = F_{XY}(x_1, y_2) - F_{XY}(x_1, y_1)$ diff of two cdf

and $P[E_3] = F_{XY}(x_2, y_1) - F_{XY}(x_1, y_1)$ diff of " "

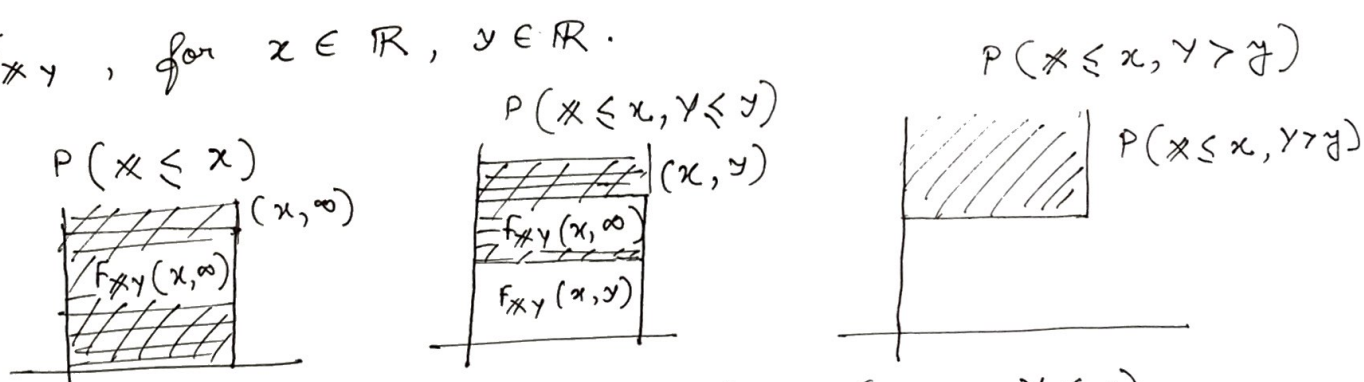
$$\therefore P(E_4) = F_{XY}(x_2, y_2) - F_{XY}(x_1, y_1) - [F_{XY}(x_1, y_2) - F_{XY}(x_1, y_1)] - [F_{XY}(x_2, y_1) - F_{XY}(x_1, y_1)]$$

$$= F_{XY}(x_2, y_2) - F_{XY}(x_1, y_2) - F_{XY}(x_2, y_1) + F_{XY}(x_1, y_1)$$

or $P(x_1 < X \leq x_2, y_1 < Y \leq y_2) = F_{XY}(x_2, y_2) - F_{XY}(x_1, y_2) - F_{XY}(x_2, y_1) + F_{XY}(x_1, y_1)$.

Ans

6. Write the prob $P(X \leq x, Y > y)$ in terms of joint cdf $F_{XY}$, for $x \in \mathbb{R}$, $y \in \mathbb{R}$.




$$P(X \leq x, Y > y) = P(X \leq x) - P(X \leq x, Y \leq y)$$

by definition,

$$P(X \le x, Y > y) = F_X(x) - F_{XY}(x, y)$$

Since,

$\{X \le x, Y \le y\}$ joint cdf & $\{X \le x, Y > y\}$ are two disjoint regions so union is $\{X \le x\}$

so $P(X \le x) = P(X \le x, Y \le y) + P(X \le x, Y > y)$ ]

Answer